

```
int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    int t; cin >> t;
    while(t--){
        int c; cin >> c;
        if(c == 1){
            int n; ll k; cin >> n >> k;
            --k;
            vector<int> temp = decimal_to_factoradix(k);
            vector<int> ans = factoradix_to_permutation(n, temp);
            for(auto &x: ans)cout << x << " ";
            cout << nl;
        } else {
            int n; cin >> n;
            vi permutation(n); for(int i=0;i<n;++i) cin >> permutation[i];
            vi factoradix = permutation_to_factoradix(n, permutation);
            ll k = factoradix_to_decimal(factoradix);
            ++k;
            cout << k << nl;
        }
    }
}
```